CCBDC
CCDCC
BCCDE
EBCAC
CDDCC
BBCBA
CB

## 4.4 monopolistic competition

**1.** An industry consists of 100 small firms, and the largest firm accounts for only 2 percent of sales. Brand names are considered a signal of quality. The industry described is best classified as

(A) monopoly
(B) perfectly competitive
(C) monopolistically competitive ✓
(D) oligopolistic
(E) monopsonistic

**2.** Which of the following is true of both monopolistically competitive and perfectly competitive firms in long-run equilibrium?

(A) Marginal revenue equals average total cost.
(B) Marginal cost equals average total cost.
(C) Price equals average total cost. ✓
(D) Price is greater than marginal cost.
(E) Production occurs at minimum average total cost.

**3.** A monopolistically competitive firm advertises in order to

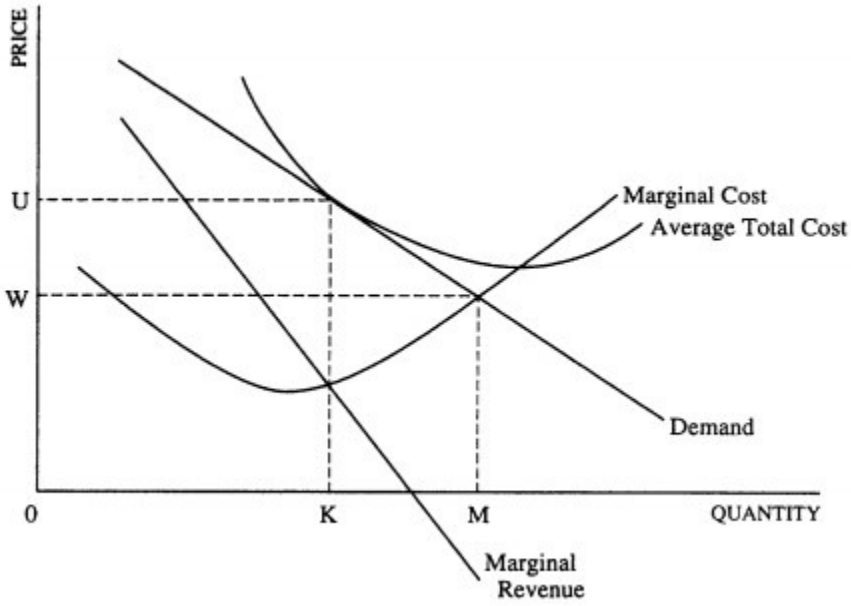
(A) shift the demand curve for its product to the left.
(B) make the demand for its product less price elastic ✓
(C) make its product more similar to its competitors'
(D) increase its positive externalities
(E) reduce the industry's barriers to entry

**4.** In monopolistic competition, a goal of advertising is to

(A) reduce a firm’s short-run average total cost
(B) minimize a firm’s long-run average total cost
(C) make a firm’s demand curve more elastic
(D) make a firm’s demand curve less elastic ✓
(E) shift a firm’s demand curve to the left

## 4.4 monopolistic competition

**5.**




The graph above depicts cost and revenue curves for a typical firm in a monopolistically competitive industry. Suppose that the firm is producing 0M units of output. To maximize profits, it should do which of the following to output and price?

## 4.4 monopolistic competition

(A)

| Output | Price |
|---|---|
| Increase | Decrease |

(B)

| Output | Price |
|---|---|
| Increase | Increase |

(C) ✓

| Output | Price |
|---|---|
| Decrease | Increase |

(D)

| Output | Price |
|---|---|
| Not change | Increase |

(E)

| Output | Price |
|---|---|
| Not change | Not change |

**6.** All of the following characterize both perfectly competitive and monopolistically competitive markets EXCEPT:

(A) Price is equal to average revenue.

(B) Individual firms produce output where marginal cost equals marginal revenue.

(C) Firms can affect the selling price of their product. ✓

(D) The market has a large number of firms.

(E) Firms can easily enter or exit the market.

**7.** A monopolistically competitive firm is currently producing the profit-maximizing level of output. If the price of a variable input increases, which of the following will occur?

## 4.4 monopolistic competition

(A) The firm will increase its output to increase its revenue.
(B) The firm will increase the price of its output by the same amount.
(C) The firm's average total cost and marginal cost curves will shift upward. ✓
(D) The firm's average fixed cost will decrease as it decreases production.
(E) The firm's fixed cost will increase, but its output level will be unaffected.

**8.** A monopolistically competitive firm's demand curve will be highly elastic if which of the following exists?

(A) A high degree of product differentiation
(B) A highly elastic supply curve for the firm
(C) High barriers to entry in this industry
(D) A high degree of product substitutability ✓
(E) A small number of competitors

**9.** A monopolistically competitive profit-maximizing firm is currently producing and selling 2,000 units of output. At this output level, marginal revenue is $9, average revenue is $10, and the average variable cost is $8. The product price is

(A) $8
(B) $9
(C) $10 ✓
(D) greater than $10
(E) less than $8

**10.** Assume that a monopolistically competitive firm is currently maximizing profit with an output of 100 units and a price of $50. Which of the following is true?

(A) Total revenue is maximized when the firm produces 100 units of output.
(B) Marginal revenue equals $50 when the firm produces 100 units of output.
(C) Marginal cost is greater than marginal revenue when the firm produces 150 units of output. ✓
(D) Marginal cost is minimized when the firm produces 100 units of output.
(E) Average total cost is minimized when the firm produces 100 units of output.

**11.** A well-known fast-food franchise substantially increases the price of its hamburgers, and loses only some of its customers. Which of the following best explains why the franchise has <u>not</u> lost all of its customers

(A) Its hamburgers are a perfect substitute for other types of fast food.
(B) Its hamburgers are differentiated. ✓
(C) The demand for its hamburgers is perfectly elastic.
(D) The other competitive fast-food restaurants decrease the price for their hamburgers.
(E) The barriers to entry are very low for entrepreneurs trying to enter the fast-food business.

## 4.4 monopolistic competition

**12.** Compared with firms in a perfectly competitive industry, firms in a monopolistically competitive industry are inefficient because they

(A) make economic profits in the long run

(B) do not lower the product price if input prices fall

(C) restrict their output level to maximize profits ✓

(D) charge the highest price that consumers will pay

(E) waste resources by producing an excess amount of output

**13.** Daily Tools is a typical firm in a monopolistically competitive market that produces tool kits for everyday use. Which of the following explains why Daily Tools is inefficient in long-run equilibrium?

(A) The entry of more efficient firms is blocked.

(B) The firm produces the quantity at which marginal cost equals average total cost.

(C) The firm does not produce the quantity that results in minimum average total cost. ✓

(D) The firm does not produce the quantity that minimizes total cost.

(E) The firm's customers value the last unit of output at less than its cost.

**Answer C**

Correct. A monopolistically competitive firm is not efficient in the long run because it does not produce at the minimum of its average cost curve. Because price is equal to average total cost for monopolistically competitive firms in long-run equilibrium, such firms will produce their profit-maximizing quantity on the downward-sloping portion of their average total cost curves. This results in a lower level of output than would occur if the firm were to produce at the minimum average total cost.

**14.** A monopolistically competitive firm's demand curve will be least elastic if

(A) the number of rival firms producing very similar products increases

(B) the number of rival firms producing more differentiated products increases

(C) the number of rival firms producing very similar products decreases

(D) the number of rival firms producing more differentiated products decreases ✓

(E) a monopolistically competitive firm's demand curve becomes perfectly elastic

## 4.4 monopolistic competition

**15.**

| Quantity | Price ($) | Total Cost ($) |
|---|---|---|
| 1 | 10 | 11 |
| 2 | 8 | 14 |
| 3 | 7 | 18 |
| 4 | 5 | 25 |
| 5 | 3 | 34 |

The table provided shows price and cost data for Howell's Toy Hoops, a typical profit-maximizing firm that sells its toys in a monopolistically competitive market. At the profit-maximizing quantity, the economic profit for Howell's Toy Hoops is

(A) -$19

(B) -$5

(C) -$1

(D) $2

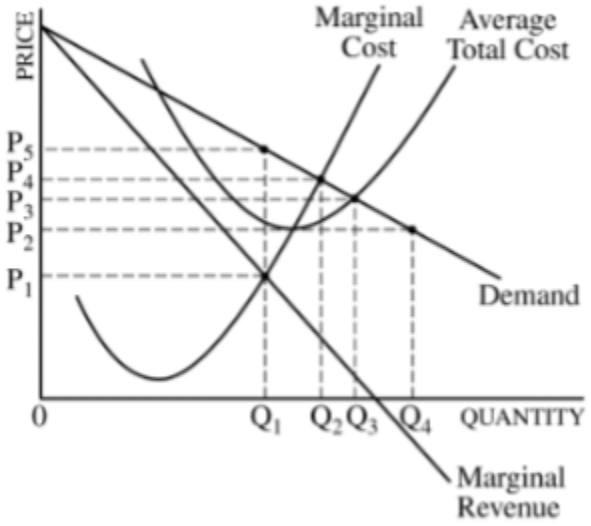
(E) $3 ✓

**Answer E**

Correct. Howell's Toy Hoops will maximize profit by producing up to the last unit where marginal revenue is greater than or equal to marginal cost. As quantity increases from 2 units to 3 units, total revenue changes from \$16 ($=2\times\$8$) at 2 units of output to \$21 ($=3\times\$7$) at 3 units of output, so marginal revenue is \$5 when 3 units of output are produced. Marginal cost is \$4 ($=\frac{\$18-\$14}{3-2}$) at 3 units of output, so this is the last unit where marginal revenue is greater than marginal cost. Economic profit at 3 units of output is total revenue (\$21) minus total cost (\$18), which equals \$3.

## 4.4 monopolistic competition

**The Question refers to the following graph, which shows the cost and revenue curves for a profit-maximizing monopolistically competitive firm.**




**16.** What is the profit-maximizing price and quantity?

## 4.4 monopolistic competition

(A)

| Price | Quantity |
|---|---|
| $P_1$ | $Q_1$ |

(B)

| Price | Quantity |
|---|---|
| $P_2$ | $Q_4$ |

(C)

| Price | Quantity |
|---|---|
| $P_3$ | $Q_3$ |

(D)

| Price | Quantity |
|---|---|
| $P_4$ | $Q_2$ |

(E) ✓

| Price | Quantity |
|---|---|
| $P_5$ | $Q_1$ |

**17.** Is the firm in short-run or long-run equilibrium?

(A) Short run, because price is greater than marginal cost

(B) Short run, because the firm is earning a positive economic profit ✓

(C) Long run, because price is greater than average total cost

(D) Long run, because marginal revenue is not equal to zero

(E) Either short run or long run, because the firm is producing where marginal revenue equals marginal cost

---

**18.** The demand curve for a monopolistically competitive firm is downward sloping because

## 4.4 monopolistic competition

(A) there are a large number of firms
(B) the product is produced by using scarce resources
(C) the products produced by different firms are not identical ✓
(D) it is easy for firms to enter or exit the market
(E) the marginal cost rises as output produced increases

**19.** Firms in monopolistic competition do not attain allocative efficiency because at the long-run equilibrium output, which of the following is true?

(A) Price is greater than marginal cost. ✓
(B) Marginal cost is greater than minimum average total cost.
(C) Marginal revenue is greater than marginal cost.
(D) There is an overallocation of resources to the market.
(E) Products are homogeneous.

**Answer A**

Correct. In long-run equilibrium, firms produce the profit-maximizing quantity by equating marginal cost to marginal revenue. The price set is greater than marginal cost, creating deadweight loss. Therefore, monopolistically competitive firms are not allocatively efficient.

**20.** In the long run, a monopolistically competitive firm is allocatively inefficient because the firm will

(A) produce only when marginal cost is greater than marginal revenue
(B) produce only when marginal revenue is greater than marginal cost
(C) charge a price greater than the marginal cost ✓
(D) earn positive economic profits
(E) experience economic losses

**21.** Let P = price, MR = marginal revenue, MC = marginal cost, and ATC = average total cost. In monopolistic competition, which of the following most accurately describes the long-run equilibrium conditions for a firm?

(A) P > ATC, MR = MC, and P > MC
(B) P > ATC, MR > MC, and P = MC
(C) P = ATC, MR = MC, and P > MC ✓
(D) P = ATC, MR = MC, and P = MC
(E) P = ATC, MR > MC, and P > MC

**22.** Which of the following is true of a monopolistically competitive firm in long-run equilibrium?

## 4.4 monopolistic competition

(A) Price is greater than marginal cost, and marginal revenue is equal to average total cost.

(B) Price is greater than marginal revenue, and marginal cost is equal to average total cost.

(C) Price is greater than marginal revenue, and marginal cost is greater than average total cost.

(D) Marginal revenue is equal to marginal cost, and price is equal to average total cost. ✓

(E) Marginal revenue is greater than marginal cost, and price is equal to average total cost.

**23.** Which of the following statements is true for both a monopolistically competitive firm and a perfectly competitive firm in long-run profit-maximizing equilibrium?

(A) Economic profits equal zero, and price equals marginal cost.

(B) Economic profits equal zero, and price equals marginal revenue.

(C) Marginal revenue equals marginal cost, and profits are positive.

(D) Economic profits equal zero, and marginal revenue equals marginal cost. ✓

(E) Economic profits equal zero, and price exceeds marginal cost.

**24.** Which of the following is true for both a monopolistically competitive firm and a perfectly competitive firm in long-run equilibrium?

(A) Marginal cost is greater than marginal revenue.

(B) Price is greater than marginal cost.

(C) Price is equal to average total cost. ✓

(D) Price is equal to marginal cost.

(E) Marginal revenue is equal to average revenue.

**25.** Which of the following statements about a monopolistically competitivefirm in long-run equilibrium is true?

(A) It has excess capacity, even though its long-run profit is zero and its output prices equals its marginal cost.

(B) It has excess capacity, and its long-run profit is positive, even though its marginal revenue equals its marginal cost.

(C) It has excess capacity and its output price exceeds its marginal cost, even though its long-run profit is zero. ✓

(D) It has no excess capacity and its long-run profit is zero.

(E) It has no excess capacity and its marginal revenue equals its marginal cost.

**26.** Monopolistically competitive firms are considered inefficient in allocating society's resources for which of the following reasons?

## 4.4 monopolistic competition

(A) In equilibrium, the marginal benefit exceeds the price charged by the firms.

(B) In equilibrium, the marginal benefit exceeds the marginal cost of production. ✓

(C) In equilibrium, the marginal revenue of the firm is not equal to its marginal cost.

(D) In long-run equilibrium, the firm is earning economic profits.

(E) Firms exhibit significant market power and therefore the number of firms in the industry is strictly limited.

**27.** Monopolistically competitive product markets are inefficient because

(A) price equals the marginal value to the buyer of the last item produced

(B) price is greater than marginal cost ✓

(C) excessive competition prevents other firms from entering the market

(D) homogeneous goods are usually overpriced

(E) short-run economic profit-making opportunities exist

**28.** Which of the following is true of a monopolistically competitive firm in long-run equilibrium?

(A) It produces where price equals marginal cost, and it earns zero economic profits.

(B) It produces where marginal revenue exceeds marginal cost, and it earns positive economic profits.

(C) It produces where marginal cost equals marginal revenue, and the price is equal to average total cost. ✓

(D) It produces at the minimum average total cost, and it utilizes all excess capacity.

(E) It produces more than the allocatively efficient quantity, and the price is greater than marginal cost.

**29.** Which of the following must be true if a profit-maximizing monopolistically competitive firm continues to operate in the short run while incurring a loss?

(A) Marginal revenue equals marginal cost and price is greater than average total cost.

(B) Marginal revenue equals marginal cost and price is greater than average variable cost. ✓

(C) Marginal revenue equals marginal cost and price is less than average variable cost.

(D) Price equals both marginal cost and marginal revenue.

(E) Price is less than marginal revenue, but greater than average variable cost.

**30.** Monopolistically competitive firms are inefficient because they

(A) produce a lower level of output at a higher average cost than do perfectly competitive firms ✓

(B) use production processes that are more capital-intensive than do perfectly competitive firms

(C) face downward-sloping demand curves, ensuring that marginal revenue is greater than average revenue

(D) produce at that level of output where price equals marginal cost

(E) realize diseconomies of scale

**31.** In long-run equilibrium, at which of the following output levels will a monopolistically competitive firm operate?

## 4.4 monopolistic competition

(A) Where long-run average total cost is at a minimum
(B) Where price equals marginal revenue
(C) Where price equals average total cost ✓
(D) Where marginal revenue equals average total cost
(E) Where marginal revenue equals average revenue

**32.** In monopolistic competition, an individual firm's market power stems from which of the following?

(A) Economies of scale in production
(B) The production of a differentiated product ✓
(C) The absence of perfectly competitive firms that produce similar products
(D) Barriers to entry allowing firms to earn economic profit
(E) The maintenance of excess capacity to meet unexpected increase in demand

**Answer B**

Correct. Monopolistic competition is a market structure that consists of many firms producing differentiated products in a market with few or no barriers to entry or exit. Product differentiation allows each firm to have market power over its differentiated product.

**33.** Which of the following is true for a monopolistically competitive firm that is in short-run equilibrium and earning a positive economic profit?

(A) Accounting profits are less than economic profits.
(B) Marginal cost is greater than price.
(C) Marginal revenue is less than price. ✓
(D) Average total cost is greater than price.
(E) Marginal revenue is less than marginal cost.

**Answer C**

Correct. Firms in monopolistic competition must lower price on all units of output to sell one more unit; marginal revenue must always be lower than price.

**34.** Which of the following is true of the marginal revenue curve for a monopolistically competitive firm?

## 4.4 monopolistic competition

(A) It is the same as the demand curve because there are many firms in the market.
(B) It is the same as the demand curve because marginal revenue is always equal to price.
(C) It is upward sloping because total revenue increases as the quantity sold increases.
(D) It lies below the demand curve because the firm must lower its price for all units in order to increase sales. ✓
(E) It lies below the demand curve because marginal revenue increases when price decreases.

**Answer D**

Correct. A monopolistically competitive firm faces a downward sloping demand curve. For the firm to sell more units of output, it must lower its price on all units. Thus, the extra revenue gained from selling the last unit, or the marginal revenue, will be less than the price charged (from the demand curve) and the marginal revenue curve will always lie below the demand curve.

**35.** Monopolistically competitive markets are characterized by

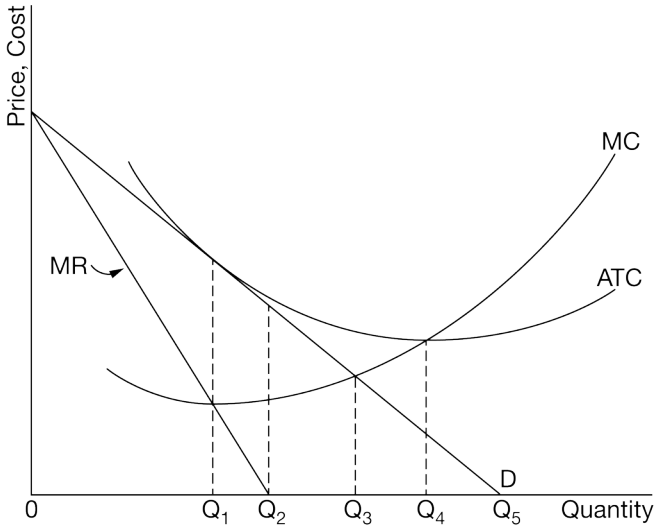
(A) a large number of firms ✓
(B) economies of scale
(C) standardized products
(D) mutual interdependence
(E) positive economic profit in the long run

**Answer A**

Correct. A large number of firms is a common characteristic in both monopolistic competition and perfect competition.

## 4.4 monopolistic competition

**36.**




For the monopolistically competitive firm represented by the graph above, the allocatively efficient quantity of output is

(A) $Q_1$
(B) $Q_2$
(C) $Q_3$ ✓
(D) $Q_4$
(E) $Q_5$

**Answer C**

Correct. $Q_3$ corresponds to where marginal cost intersects demand which is the allocatively efficient quantity.

**37.** Which of the following is true of monopolistically competitive firms in long-run equilibrium?

(A) Firms can earn positive economic profits.
(B) Firms face a perfectly elastic demand curve.
(C) Price equals marginal cost, which equals marginal revenue.
(D) Price will always be above average total cost.
(E) Marginal revenue equals marginal cost, and price equals average total cost. ✓

## 4.4 monopolistic competition

**Answer E**

Correct. The profit-maximizing firm produces where marginal revenue equals marginal cost. With easy exit and entry, in the long run, firms earn zero economic profit in monopolistic competition, and therefore, price equals average total cost.

**38.** Which of the following is true about price $(\mathrm{P})$, marginal cost $(\mathrm{MC})$, marginal revenue $(\mathrm{MR})$, and economic profit of a monopolistically competitive firm in long-run equilibrium?

(A) $\mathrm{P} > \mathrm{MC}$; Marginal revenue is below demand; Economic profit is zero ✓

(B) $\mathrm{P} < \mathrm{MC}$; Marginal revenue is below demand; Economic profit is positive

(C) $\mathrm{P} > \mathrm{MC}$; Marginal revenue is above demand; Economic profit is positive

(D) $\mathrm{P} = \mathrm{MR} = \mathrm{MC}$; Marginal revenue is equal to demand; Economic profit is zero

(E) $\mathrm{P} = \mathrm{MR} = \mathrm{MC}$; Marginal revenue is above demand; Economic profit is positive

**Answer A**

Correct. A monopolistically competitive firm is a price maker and always charges a price above its marginal cost. Since the firm faces a downward sloping demand curve, it must lower its price on all units of output to sell more units, and therefore its marginal revenue curve will always lie below the demand curve. With low barriers to entry and exit in monopolistic competitive markets, firms earn zero economic profit in the long run.

**39.** Which of the following is true for a firm in long-run equilibrium in monopolistic competition?

(A) Given barriers to entry, the firm earns economic profits in long-run equilibrium.

(B) The firm is productively efficient, producing at the minimum of long-run average total cost.

(C) Price equals marginal revenue and marginal cost.

(D) There is neither allocative nor productive efficiency. ✓

(E) Price is greater than average total cost in long-run equilibrium.

**Answer D**

Correct. With easy exit and entry, in the long run, firms earn zero economic profit in monopolistic competition. The firm faces a downward-sloping demand curve, so price is greater than marginal cost,

## 4.4 monopolistic competition

which does equal marginal revenue at the profit-maximizing output level. With price greater than marginal cost, there is no allocative efficiency. Also, since average total cost is falling, where the firm is producing, it is not at the minimum of long-run average cost, so there is no productive efficiency.

**40.** Which of the following is true of a monopolistically competitive firm in long-run equilibrium?

(A) Price exceeds marginal revenue, and the firm earns positive economic profits.

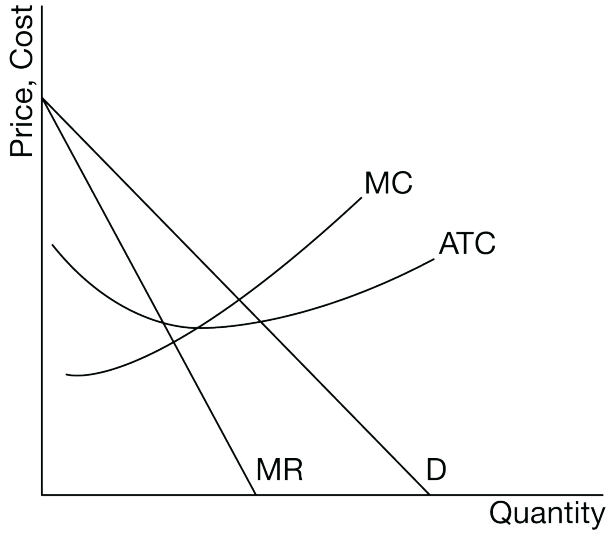
(B) Price equals marginal revenue, and the firm earns positive economic profits.

(C) Price equals marginal cost, and the firm earns zero economic profits.

(D) Price exceeds marginal cost, and the firm earns zero economic profits. ✓

(E) Price exceeds marginal cost, and the firm earns positive economic profits.

**41.**




Based on the information in the above graph describing a monopolistically competitive firm, which of the following is true?

(A) The firm is productively efficient since marginal cost intersects average cost at the minimum of average cost.

(B) When maximizing profits, this firm will have economic losses but will still continue to produce.

(C) The firm will produce where marginal revenue equals marginal cost and will set its price to equal average total cost.

(D) With the firm making economic profits, it can be expected that new firms will enter this market. ✓

(E) The firm will produce where demand is inelastic to capture all economies of scale.

## 4.4 monopolistic competition

**Answer D**

Correct. The profit-maximizing firm produces where marginal revenue equals marginal cost and, from the demand curve above that output level, derives its product price. In this case, that price will be above average total cost and the firm will make economic profits. With profits being earned, new firms will enter the market over time.

**42.** Which of the following is true for a monopolistically competitive firm in long-run equilibrium?

(A) It earns a normal profit. ✓
(B) It sets price equal to marginal cost.
(C) It is allocatively efficient.
(D) It is a price taker.
(E) It produces at minimum average total cost.

**43.** Which of the following is NOT a characteristic of monopolistically competitive markets?

(A) Relatively easy market entry
(B) Differentiated products
(C) Substantial product advertising
(D) A large number of both buyers and sellers
(E) Long-run economic profits ✓

**44.** Which of the following statements correctly identifies a difference between perfect competition and monopolistic competition?

(A) In perfect competition there are no barriers to entry, but there are strong barriers in monopolistic competition.
(B) In perfect competition there are many firms, but in monopolistic competition there are only a few firms.
(C) In perfect competition the firms all sell products that are exactly the same, but in monopolistic competition each firm sells a slightly differentiated product. ✓
(D) In perfect competition firms maximize profit by selling the quantity where marginal revenue equals marginal cost, but in monopolistic competition firms maximize profit by selling the quantity where marginal revenue exceeds marginal cost.
(E) In perfect competition there are few consumers, but in monopolistic competition there are many consumers.

**45.** Which of the following about the relationship between marginal revenue (MR) and price (P) under monopolistic competition and perfect competition is correct?

## 4.4 monopolistic competition

(A)

| Monopolistic Competition | Perfect Competition |
|---|---|
| MR>P | MR=P |

(B) ✓

| Monopolistic Competition | Perfect Competition |
|---|---|
| MR<P | MR=P |

(C)

| Monopolistic Competition | Perfect Competition |
|---|---|
| MR decreases when P increases | MR increases when P decreases |

(D)

| Monopolistic Competition | Perfect Competition |
|---|---|
| MR=P | MR=P |

(E)

| Monopolistic Competition | Perfect Competition |
|---|---|
| MR=P | MR<P |

**46.** Over the past 5 years, 50 new restaurants have opened and 30 have closed in the city of Zuni. Currently there are 110 restaurants operating in the city. Which of the following best represents the market structure, barriers to entry, and economic profits in the long run?

## 4.4 monopolistic competition

(A)

| Market Structure | Barriers to Entry | Long-Run Economic Profits |
|---|---|---|
| Perfect competition | Low | Negative |

(B)

| Market Structure | Barriers to Entry | Long-Run Economic Profits |
|---|---|---|
| Perfect competition | High | Positive |

(C) ✓

| Market Structure | Barriers to Entry | Long-Run Economic Profits |
|---|---|---|
| Monopolistic competition | Low | Zero |

(D)

| Market Structure | Barriers to Entry | Long-Run Economic Profits |
|---|---|---|
| Monopolistic competition | High | Positive |

(E)

| Market Structure | Barriers to Entry | Long-Run Economic Profits |
|---|---|---|
| Monopolistic competition | High | Zero |

**47.** Which of the following is true of a monopolistically competitive firm in long-run equilibrium?

(A) The firm produces the allocatively efficient level of output.

(B) The firm is allocatively inefficient, because it produces an output level at which price is greater than marginal cost. ✓

(C) The firm produces an output level that minimizes average total cost.

(D) The firm produces in the inelastic range of its demand curve.

(E) The firm earns positive economic profits but zero accounting profits.

**48.** Which of the following is true of a monopolisti-cally competitive firm in long-run equilibrium?

## 4.4 monopolistic competition

(A) Price equals marginal cost and average total cost.

(B) Price equals average total cost but is greater than marginal cost. ✓

(C) Price equals marginal cost and is greater than average total cost.

(D) The firm makes no attempt to differentiate its products.

(E) The firm earns positive economic profits by producing at minimum average cost.